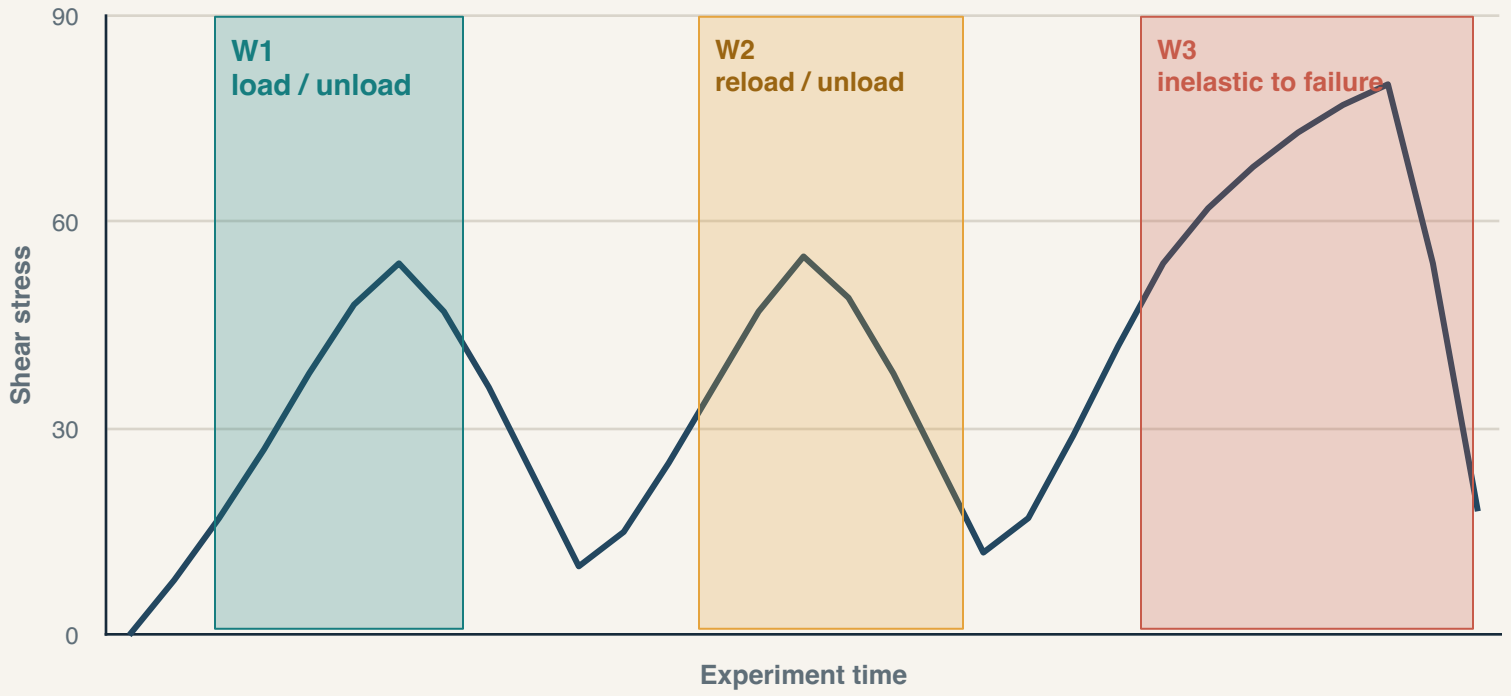
UCS + AE laboratory protocol

Cyclic loading with three synchronized waveform-recording windows

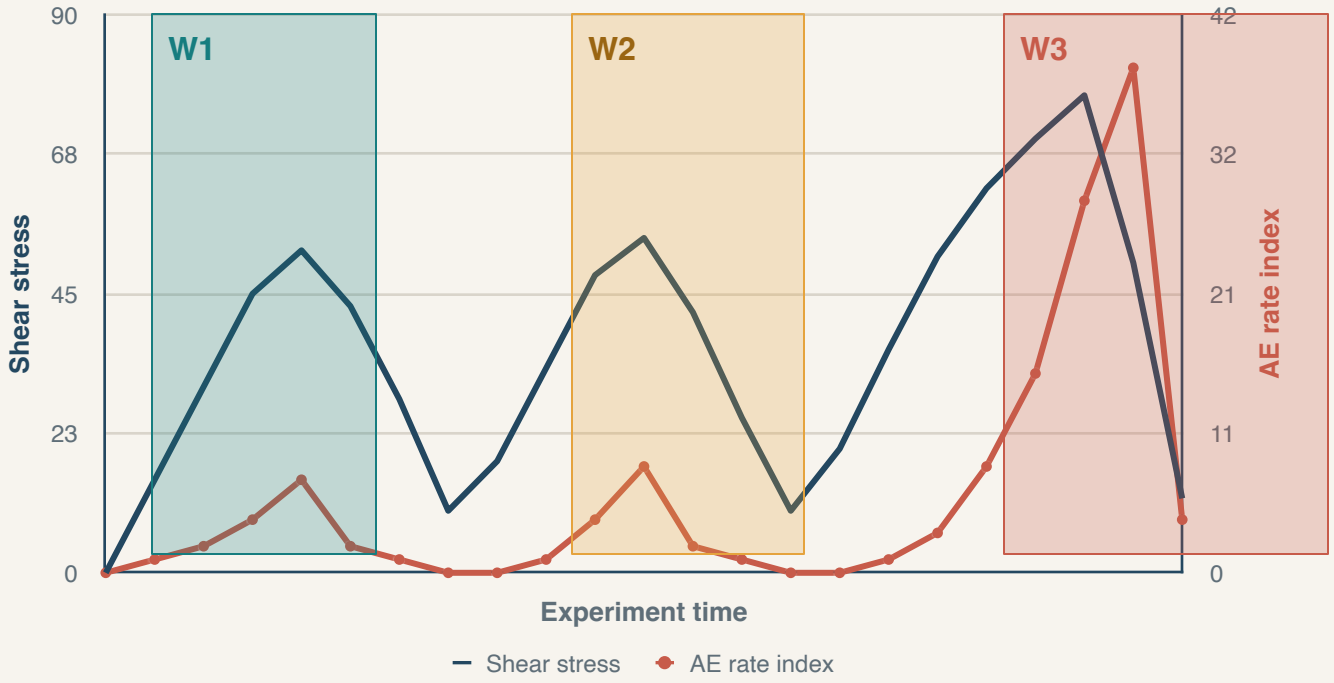


Acquisition sequence

1. **Mechanical loading**
   Apply loading, unloading and reloading cycles.
2. **Synchronization**
   A pulse in the mechanical file marks each AE-window onset.
3. **Waveform acquisition**
   Record every channel at 3.25 MHz in consecutive 1 s chunks.
4. **Event catalogue**
   Detect events and map time and amplitude onto the mechanical record.

Why record three targeted windows?

The protocol compares repeat loading with the transition from distributed damage to failure



Questions addressed

Memory during reloading

Does AE activity remain suppressed until stress approaches the previous maximum?

Onset of inelastic damage

Do event rate and amplitude change when the mechanical response departs from elastic behavior?

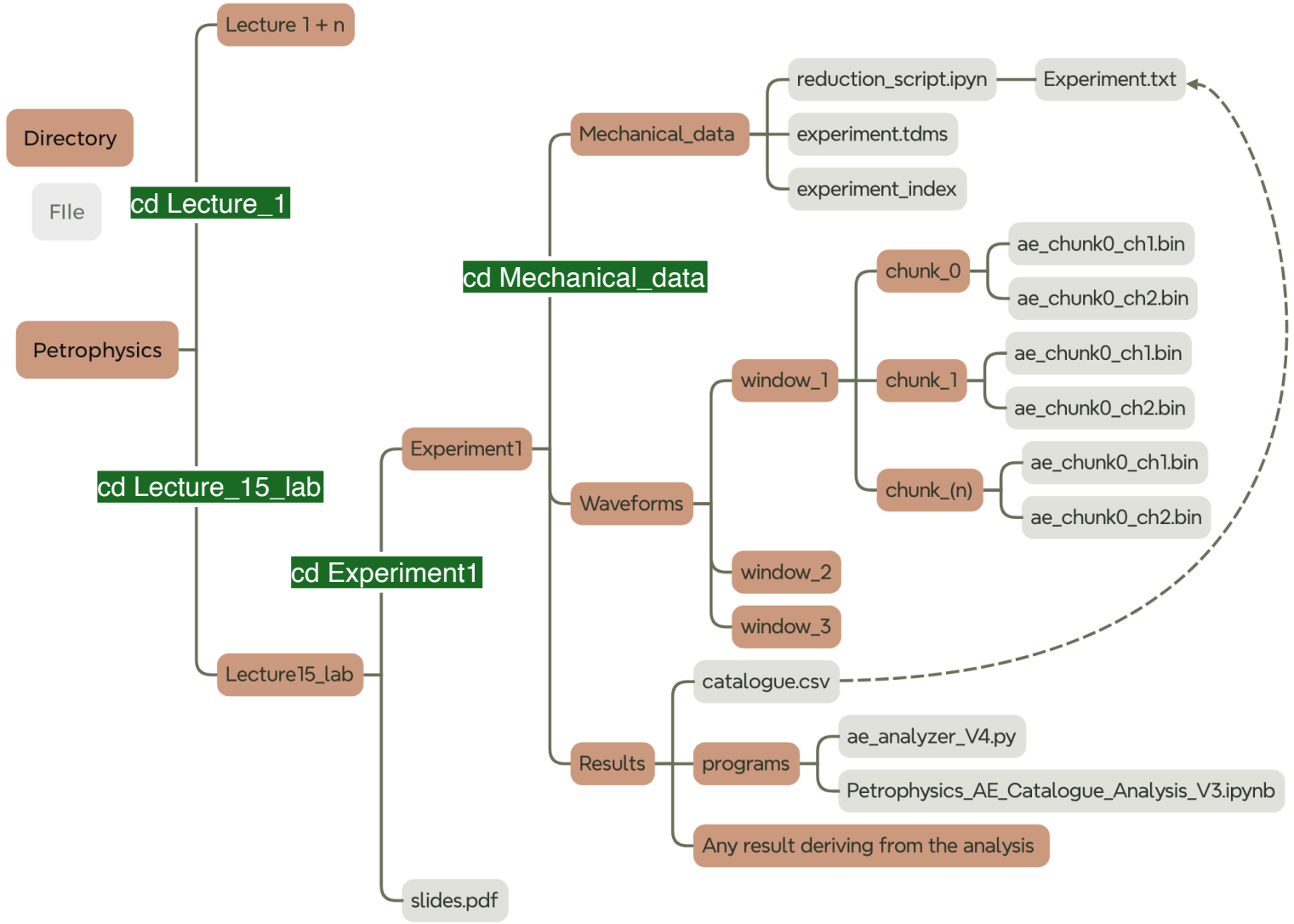
Approach to failure

Does activity accelerate or shift toward larger amplitudes before macroscopic failure?

Interpretation limit

Selective recording controls data volume, but gaps are unobserved intervals. They cannot be interpreted as zero AE activity.

Data structure



Main directories

Mechanical data

We store all the informations about the experiment and the files related to the mechanical data.

Waveforms directory

Here we put all the data related to the raw data waveforms to be analysed

Results directory (and subdirectories)

Here we store all the codes for the data analysis, the results from each analysis.

How to read the diagram

Brown boxes — directories (folders)

Grey boxes — files

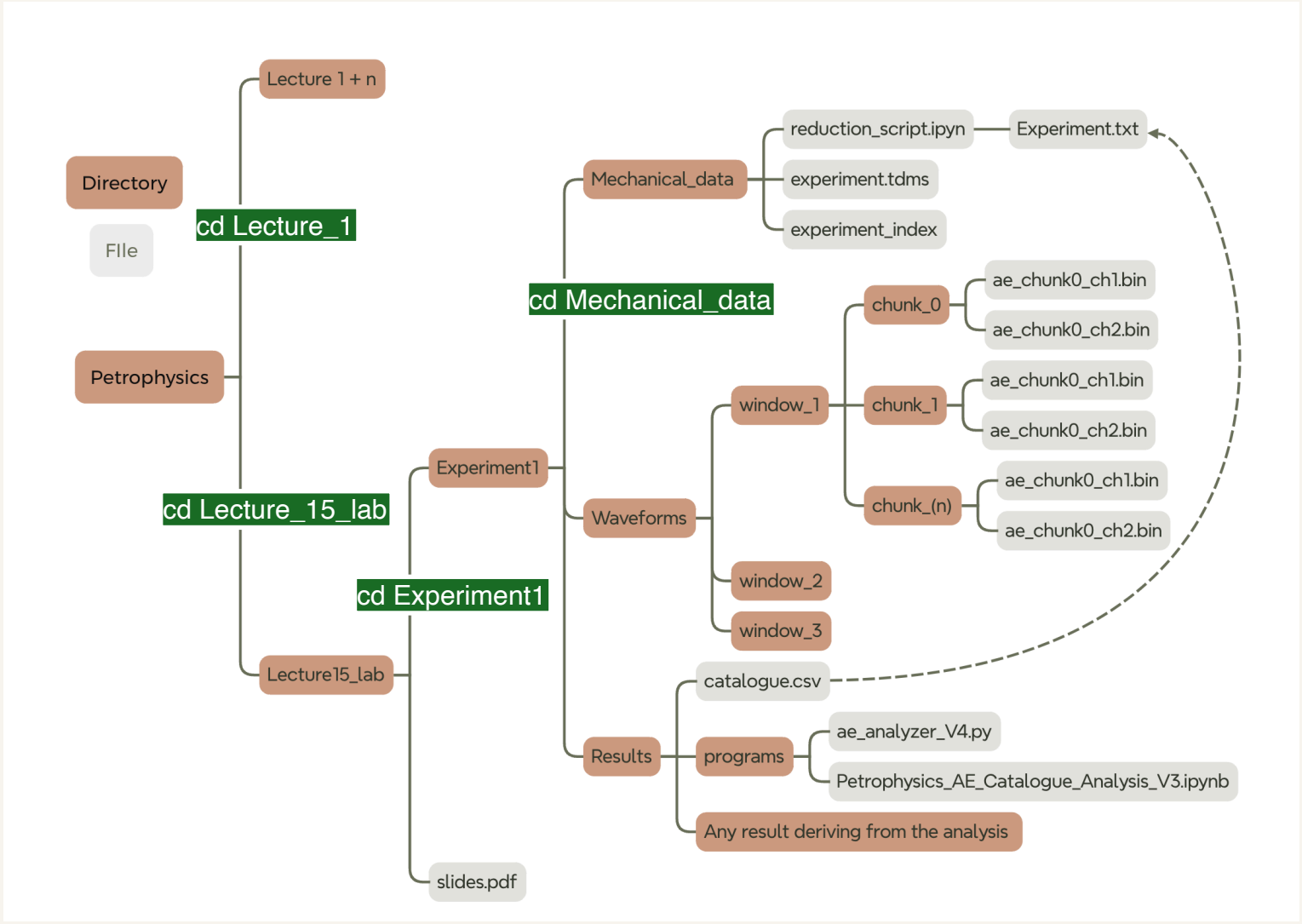
Move into a folder

`cd folder_name`

Return to the parent folder

`cd ..`

Data structure - navigation



Directory = folder

A folder can contain files and other folders.
A path is the address of one item in this tree.

CURRENT WORKING DIRECTORY

…/Experiment_1/Results/programs

Relative file paths start from this location.

NAVIGATING IN A TERMINAL

`pwd`	show the current directory
`ls`	list files and folders here
`cd Results`	enter the Results folder
`cd ..`	return to the parent folder

In Jupyter, the file browser performs the same navigation by clicking folders and using the back button.

Data structure and directory navigation - try this on a terminal or anaconda prompt

Folders form a tree. Terminal commands move the current working directory through that tree.

